

Anchor Story:

- “Develop a function to determine all spaces that have mines.”
 - Story Points: 1
 - Duration: 15 minutes
 - Used as the basis for estimating Project 2 User Stories, using Project 1 reference stories as a guide when applicable.
-
1. “As a player, I want to compete with an AI opponent in an interactive turn-based minesweeper match at various difficulties, so I can be challenged.”
 - a. 5 story points.
 - i. **1.25 hours**
 - b. Reference Project 1 User Story: 2.1-2.4
 - i. This user story will require integrating new components with existing code in order to preserve the functionality implemented by these user stories. It will require understanding how these stories are implemented, and how the existing code can be used.
 - ii. Collectively took 1.75 hours(8 story points).
 - c. Rationale: defining class, initial designs, method definitions, integration with existing codebase could be tricky.
 2. “As an inexperienced player, I want my AI opponent that randomly selects covered cells, so the game is fairer.”
 - a. 3 story points
 - i. **0.75 hours**
 - b. Reference Project 1 User Story: 2.1-2.2
 - i. These user stories collectively involved working directly with grid cells and their attributes, such as the presence of a mine. Similar activities will be needed for this user story.
 - ii. Collectively took 0.75 hours(4 story points) last time.
 - c. Rationale: already would have an existing class for AI, would only need to randomly select covered cells.
 3. “As a moderately skilled player, I want my AI opponent to strategically select cells once a safe cell is uncovered, so I have more of a challenge.”

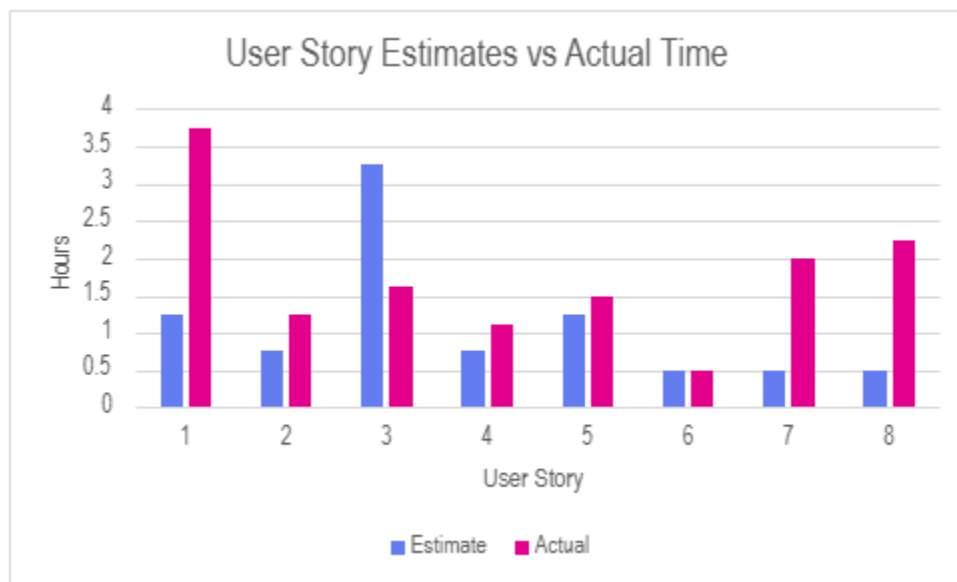
- a. 13 story points
 - i. **3.25 hours**
 - b. Reference Project 1 User Story: 2.1-2.4
 - i. These combined user stories required developing logic that is comparable to the amount of logic that would be needed for a medium difficulty mode, such as checking adjacent mine counts for uncovered cells, discerning obvious mine cells, and other strategic steps.
 - ii. These reference stories are used as a starting point, since effective strategy derivation was not needed in the previous project, and may require more effort.
 - iii. Took a combined 1.75 hours(8 story points) last time.
 - c. Rationale: Requires devising an algorithm to solve minesweeper may be more complicated than foreseen.
4. “As an advanced player, I want my AI opponent to only select cells that are certain to have no mines, so I have the maximum challenge.”
- a. 3 story points
 - i. **0.75 hours**
 - b. Reference Project 1 User Story: 2.1-2.2
 - i. These user stories collectively involved working directly with grid cells and their attributes, such as the presence of a mine. Similar activities will be needed for this user story.
 - ii. Comparable level of effort for user story 2, but with an additional step.
 - iii. Collectively took 0.75(4 story points) hours last time.
 - c. Rationale: Requires obtaining all spaces with mines (already decided to be 1 story point). Also requires random selection. Comparable to user story 2.
5. “As a player, I want the option to make the AI play minesweeper automatically using any difficulty, so I can learn effective strategies.”
- a. 5 story points
 - i. **1.25 hours**
 - b. Reference Project 1 User Story: 2.1-2.4

- i. Will require integration with existing code for move implementation and status checking, and may also require modifications that preserve functionality implemented by these user stories.
 - ii. Collectively took 1.75 hours last time.
 - c. Rationale: must integrate automated mode in a codebase that seems to rely on user input.
- 6. “As a player, I want the option to choose between interactive mode and automatic mode with my AI opponent in the user interface.”
 - a. 2 story points
 - i. **0.5 hours**
 - b. Reference Project 1 User Story: 5.1 and 5.2
 - i. These stories depicted user interface additions, which this user story will require.
 - ii. It took a combined 0.68 hours(2 story points) last time.
 - c. Rationale: Simpler than anchor story, but none of us have used the button before.
- 7. “As a player, I would like to see the time it took me to win minesweeper compared against the lowest recorded time, so I can track my development.”
 - a. 2 story points.
 - i. **0.5 hours**
 - b. Rationale: Simple procedure (based on previous experience), but will have to integrate it with existing code.
 - c. Reference Project 1 User Story: 4.2
 - i. Depicts an identical timing mechanism.
 - ii. It took around 2 hours (8 story points) in actuality.
- 8. “As a player, I would like there to be sound effects for my actions, to make the experience more engaging.”
 - a. 2 story points.
 - i. **0.5 hours**
 - b. Reference Project 1 User Story: 2.5
 - i. Like 2.5, this user story seeks to add bonus effects to existing code actions. It should require minimal interference with existing code. The

only difference between this and user story 2.5 is the format of the special effects.

ii. Took 0.1 hours(2 story points) last time.

Estimated Time vs Actual Time



This time, most of our estimates were closer to the actual time required to implement the user stories. However, there is still a broad trend of underestimating the amount of time required to implement, as for all but User Story 3 and 6, we underestimated the amount of time. In the future, we will need to discuss as a team why we consistently underestimate the task duration, and how we can avoid this in the future.